

BISMARCK, ND, USAWMO: **727640**Lat: **46.783N**Lon: **100.757W**Elev: **503**StdP: **95.42**Time Zone: **-6.00 (NAC)**Period: **94-19**WBAN: **24011****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-27.5	-24.4	-31.1	0.2	-27.1	-28.0	0.3	-24.0	13.5	-6.8	12.1	-8.5	3.0	320	0.661

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	13.8	33.9	21.3	31.9	20.7	30.1	20.0	23.7	30.0	22.4	28.9	21.3	28.0	5.4	160

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
21.7	17.4	27.8	20.3	16.0	26.1	19.1	14.8	24.5	73.8	29.7	68.6	28.9	64.2	28.1	28.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 12.0	10.7	9.2	-32.5	38.4	3.6	2.5	-35.1	40.2	-37.3	41.7	-39.3	43.1	-41.9	44.9		
(5)			-32.1	25.5	2.8	1.4	-34.2	26.5	-35.8	27.3	-37.4	28.0	-39.4	29.0		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	6.4	-10.6	-8.3	-1.3	6.3	13.0	18.6	22.2	21.0	15.7	7.5	-0.9	-7.8
	DBStd	12.81	8.07	7.68	7.51	5.69	4.68	3.81	3.30	3.50	4.71	5.09	6.25	7.12
	HDD10.0	2658	638	512	353	134	24	1	0	0	7	109	327	553
	HDD18.3	4674	896	746	609	360	174	41	6	14	104	337	576	811
	CDD10.0	1334	0	0	3	24	117	260	378	341	179	31	1	0
	CDD18.3	308	0	0	0	1	8	51	126	97	26	1	0	0
	CDH23.3	3753	0	0	1	26	159	597	1423	1129	383	36	0	0
	CDH26.7	1464	0	0	0	5	39	185	609	478	139	8	0	0
Wind	WSAvg	4.1	4.0	4.1	4.5	4.8	4.7	4.2	3.8	3.8	4.0	4.2	3.9	3.9
Precipitation	PrecAvg	435	12	12	21	38	60	78	65	54	39	28	15	14
	PrecMax	685	57	44	81	139	187	202	349	236	176	120	60	53
	PrecMin	258	1	1	2	3	6	13	5	0	1	1	0	0
	PrecStd	112	9	9	15	29	42	41	53	46	33	27	14	9
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	8.8	11.5	21.4	26.7	30.5	33.4	37.6	37.0	34.1	27.5	18.8	10.1
		MCWB	4.2	5.4	10.8	14.1	18.2	21.7	21.7	21.8	20.0	15.7	10.6	4.7
	2%	DB	5.3	7.1	16.3	22.5	27.4	30.7	34.2	33.7	30.3	22.7	14.2	6.7
		MCWB	2.3	3.2	8.8	12.3	16.8	20.6	21.9	20.8	18.8	13.6	7.5	2.9
	5%	DB	3.1	4.2	12.3	19.5	24.8	28.7	32.0	31.5	27.7	19.5	10.9	3.9
		MCWB	1.0	1.7	6.8	10.4	15.2	19.3	21.4	20.4	17.5	11.7	5.7	1.2
	10%	DB	0.9	2.2	8.9	16.7	22.3	26.8	30.1	29.3	25.0	16.2	7.8	1.6
		MCWB	-0.6	0.4	4.8	9.0	14.1	18.1	20.7	19.5	16.3	10.3	3.9	-0.5
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	4.3	5.8	11.9	15.2	20.2	24.5	25.8	24.9	21.9	17.1	10.8	5.2
		MCDB	8.4	10.8	19.5	24.4	27.5	30.6	31.5	30.9	29.5	24.5	17.8	9.4
	2%	WB	2.4	3.5	9.2	12.7	18.4	22.3	24.2	22.9	20.4	14.6	8.0	3.1
		MCDB	5.0	6.5	15.5	21.2	25.3	28.2	30.6	29.4	27.2	20.8	13.5	6.4
	5%	WB	1.1	1.8	6.9	11.1	16.5	20.6	23.0	21.7	18.9	12.5	6.0	1.3
		MCDB	3.0	3.9	12.1	18.1	22.7	27.1	29.3	29.0	25.1	18.3	10.5	3.8
	10%	WB	-0.5	0.5	4.9	9.5	15.0	19.4	21.8	20.6	17.4	10.7	4.1	-0.3
		MCDB	0.9	2.2	8.7	16.0	20.8	25.3	28.1	27.4	23.6	15.7	7.6	1.6
Mean Daily Temperature Range	5% DB	MDBR	10.5	10.6	11.3	13.2	13.2	12.7	13.8	14.4	14.1	12.4	11.2	9.9
		MCDBR	12.7	13.0	17.4	19.0	17.3	15.9	17.1	17.8	18.7	19.0	16.6	13.5
		MCWBR	10.2	9.7	10.7	10.1	8.6	7.5	7.0	7.2	8.4	10.6	10.6	10.2
	5% WB	MCDBR	12.5	12.7	16.9	17.5	15.1	14.0	14.2	15.1	15.4	17.2	16.1	13.5
		MCWBR	10.1	9.6	10.7	9.7	8.2	7.4	7.1	7.3	7.7	10.0	10.5	10.4
Clear-Sky Solar Irradiance	taub	0.264	0.270	0.294	0.333	0.355	0.376	0.382	0.389	0.345	0.317	0.282	0.261	
	taud	2.430	2.442	2.462	2.381	2.390	2.374	2.385	2.339	2.443	2.484	2.472	2.451	
	Ebn at Noon	847	916	939	923	909	889	877	855	869	845	814	808	
	Edh at Noon	67	81	93	112	115	118	115	116	95	78	64	58	
All-Sky Solar Radiation	RadAvg	1.65	2.70	3.77	4.92	5.76	6.43	6.69	5.56	4.26	2.68	1.69	1.23	
	RadStd	0.14	0.26	0.29	0.33	0.49	0.48	0.31	0.41	0.30	0.32	0.24	0.09	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (10 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page